Project Report: Sinner–Alcaraz on Bluesky: A Social Network & Sentiment Analysis of the 2025 US Open

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Abstract

This report explores how the tennis rivalry between **Jannik Sinner** and **Carlos Alcaraz** shaped the online discourse on the **Bluesky** social platform during the **2025 US Open**. Gathering a dataset of **9,078 posts** via a day-by-day crawling strategy, we reconstruct the network of replies and mentions to map how communities formed, while tracing the emotional and sentiment dynamics triggered by match outcomes. The analysis reveals a highly modular network where fans cluster around shared news hubs rather than forming isolated echo chambers. Furthermore, by integrating neural sentiment signals with user interaction frequencies, we demonstrate how a sentiment-aware stance classifier can successfully refine and correct the classification errors inherent in simpler, frequency-only baselines.

Contents

1 Research Question

The goal of this project is to **characterise the online discourse** surrounding two of tennis's biggest rivals—Jannik Sinner and Carlos Alcaraz—on the Bluesky social platform during the 2025 US Open. In doing so, this study is guided by the following overarching **research question**:

"How do the Jannik Sinner and Carlos Alcaraz fan communities on Bluesky manifest at both a structural-topological and emotional level during a Grand Slam tournament like the US Open?"

To address this research question, we build a modular, reproducible NLP and social-network analysis pipeline. Concretely, the analysis aims to:

- **Describe** the social-network structure built from replies and mentions, identify cohesive communities, and measure user influence via centrality metrics.
- **Quantify** sentiment and emotion expressed in posts using state-of-the-art NLP models (RoBERTa, NRC lexicon, GoEmotions) and compare two emotion backends.
- **Assign a stance** (pro-Sinner, pro-Alcaraz or neutral) to each user by fusing sentiment signals with mention frequency, and evaluate how this refined stance differs from a simpler frequency-only baseline.
- **Visualise** all results to support interpretability—network graphs, sentiment time series, emotion profiles and stance distributions.

2 Data

2.1 Collection strategy

Data were collected from Bluesky using its official `atproto` API (`src/crawler.py`). Because the search endpoint does not guarantee exhaustive retrieval over a large window, a **day-by-day** crawling strategy was used:

- For each 24-hour window from `2025-08-21` to `2025-09-10` (UTC), two keyword queries were issued—one **Sinner-oriented** and one **Alcaraz-oriented**—each capped at **5,000 posts/day** to respect rate limits while capturing peak tournament activity.
- Results from both queries were merged and deduplicated by post URI.

2.2 Dataset description

After crawling and deduplication the dataset contained **9,078 posts**. The raw schema (`data/sinner_alcaraz_posts.csv`) is summarised in Table 1.

Field	Description
`created_at`	UTC timestamp
`uri`	Unique post identifier
`text`	Original post text
`author_handle`	User's handle
`author_did`	Decentralised identifier
`reply_count`, `like_count`, `repost_count`	Engagement metrics
`mentions`	List of mentioned user DIDs
`links`	URLs in the post
`reply_parent_did`	DID of the user being replied to
`query_source`	Which query found the post (sinner/alcaraz)

Table 1: Raw data fields collected from Bluesky.

2.3 Preprocessing & enrichment

The raw CSV goes through a preprocessing phase to be transformed into an enriched dataset (`data/sinner_alcaraz_processed.csv`, 44 columns) that serves as the single source of truth for all downstream stages. The enrichment (`src/preprocessing.py`) includes:

- **Text cleaning** (HTML unescape, URL/handle removal, normalisation) and **lemmatisation** (spaCy + TweetTokenizer).
- **RoBERTa sentiment** via `cardiffnlp/twitter-roberta-base-sentiment-latest`, producing a `sentiment_category` (positive/neutral/negative) and a continuous `sentiment_compound` $= P(\text{pos}) - P(\text{neg})$.
- **Emotion** with two backends: the **NRC lexicon** (NRCLex) over the 8 Plutchik emotions, and **GoEmotions** (`SamLowe/roberta-base-go_emotions`), whose 28 fine-grained labels are collapsed onto the same 8 categories for comparison.
- **Named entity recognition** (spaCy) and **entity linking** to detect mentions of Sinner or Alcaraz.
- **Post- and user-level stance** (see Section 3.2).

3 Methodology

The analytical pipeline is a strict DAG (orchestrated by `main.py`).

3.1 Network construction

Two graphs are built:

- an undirected graph G_u, where the edge weight is the number of interactions between users;
- a directed graph G_d that captures the flow of communication from an author to the exact user they are replying to or mentioning.

For both, we compute degree, closeness and betweenness centrality, in-/out-degree, and PageRank.

To understand how these users naturally cluster together, we apply the Louvain community detection algorithm to the undirected graph, while using the Infomap algorithm to detect communities within the directed graph.

Filtered subgraphs (we selected the top 5 largest communities) are used for visualisation and stance analysis.

3.2 Stance detection

Post-level stance assigns the RoBERTa compound score to the mentioned player for single-mention posts; dual-mention posts are split via sentence-level sentiment. Similarly, post-level emotion stance assigns the net GoEmotions (BERT) emotion valence (the sum of positive emotions: joy, trust, and anticipation, minus the sum of negative emotions: anger, disgust, sadness, and fear) to the mentioned player. User-level net stance fuses mean sentiment, mention frequency, and mean emotion stance:

$$\mathrm{net\_stance} = 0.50\,(\mathrm{mean_s} - \mathrm{mean_a}) + 0.15\,(\mathrm{freq_s} - \mathrm{freq_a}) + 0.35\,(\mathrm{mean\_emo_{sinner}} - \mathrm{mean\_emo_{alcaraz}})$$

Users are labelled `sinner` / `alcaraz` / `neutral` with a ± 0.05 threshold. Polarisation is measured via stance assortativity, the cross-stance edge ratio, and the standard deviation of net stance.

4 Results

4.1 Network structure & communities

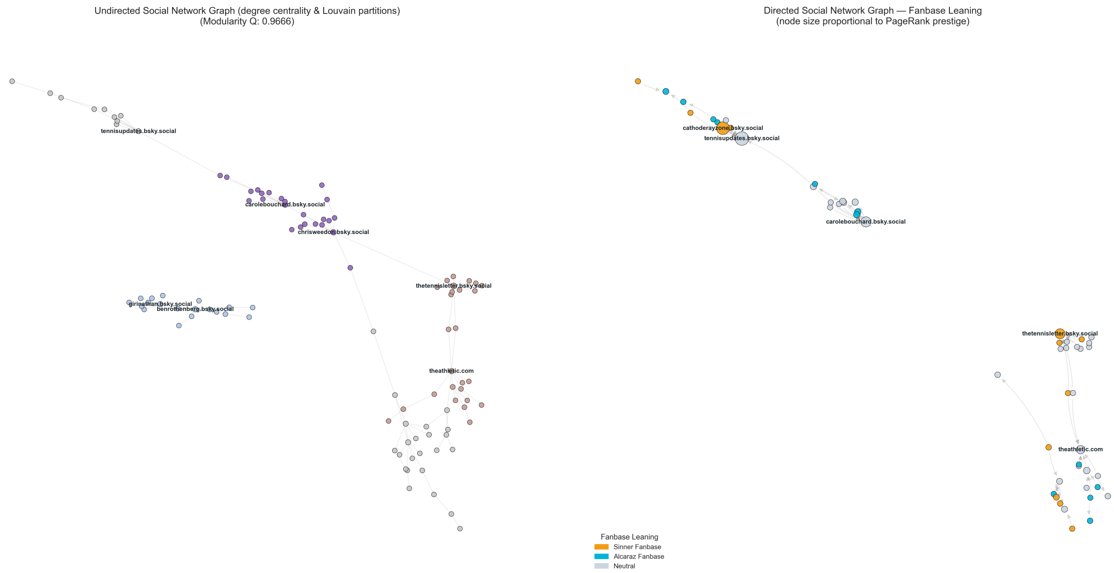
The interaction graph is **sparse but strongly modular**. Global metrics (Table 2) report a density of ≈ 0.002 and very high community modularity (Louvain ≈ 0.97, Infomap ≈ 0.94): fans cluster tightly into fanbase-aligned groups rather than mixing. The giant connected component spans only **111 users ($\approx$16.3%)**, indicating that much of the conversation happens in small, loosely linked clusters.

Metric	Value
Density	0.00203
Transitivity	0.02022
Average clustering	0.01120
Giant connected component (GCC) size	111
GCC fraction of users	0.1625
Louvain modularity	0.9666
Infomap modularity	0.9437

Table 2: Graph-level metrics (`data/network_global_metrics.csv`).

Figure 1 shows the filtered undirected network coloured by fanbase leaning, with node size proportional to degree centrality. High-degree hubs are predominantly news/aggregator accounts (e.g. `tennisupdates`, `theathletic.com`, `thetennisletter`), around which smaller fan clusters of mixed Sinner/Alcaraz/neutral leaning form. The same topology coloured by community and by dominant emotion is shown in Figure 2.

Figure 1: Filtered undirected interaction network coloured by fanbase leaning (orange = Sinner, blue = Alcaraz, grey = neutral); node size $\propto$ degree centrality. Mixed-leaning clusters form around large aggregator/news hubs.



(a) Coloured by Louvain community.

(b) Directed graph, coloured by fanbase.

Figure 2: Filtered network views. Left: community structure on G_u. Right: the directed reply/mention graph G_d coloured by fanbase leaning. Full-resolution and emotion-coloured variants are also generated in `plots/` and `plots/filtered/`.

4.2 Sentiment & emotion profiles

Overall sentiment. The RoBERTa classifier labels the bulk of posts as **neutral (77.2%)**, with **positive (16.2%)** clearly outweighing **negative (6.6%)** (Figure ??). Among posts that do carry sentiment, the conversation skews positive—tennis fans express encouragement and praise far more often than hostility.

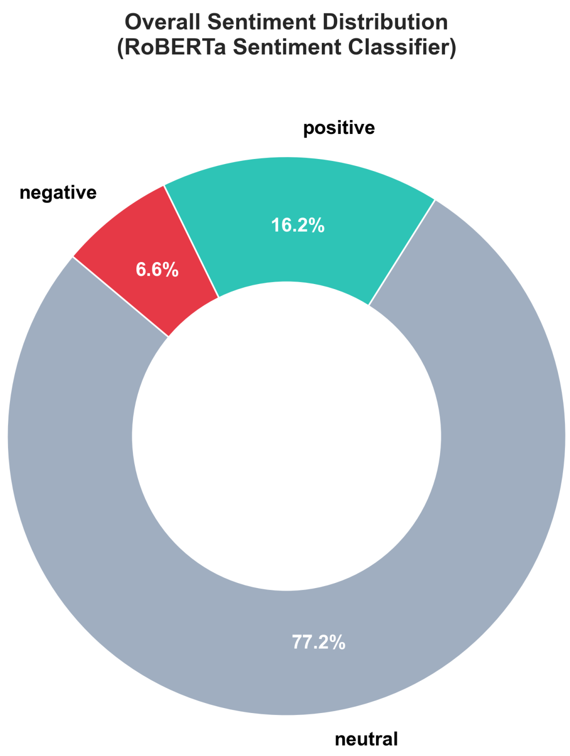


Figure 3: Overall sentiment distribution (RoBERTa). Neutral dominates; positive posts outnumber negative roughly 2.5:1.

Sentiment over time. Figure **??** plots daily average compound sentiment for each fanbase against post volume, annotated with tournament milestones. Sentiment is volatile and reacts to match outcomes: both fanbases swing around the rounds, and the lines **diverge around the final** (7 September, won by Alcaraz), while post volume builds towards the closing weekend.

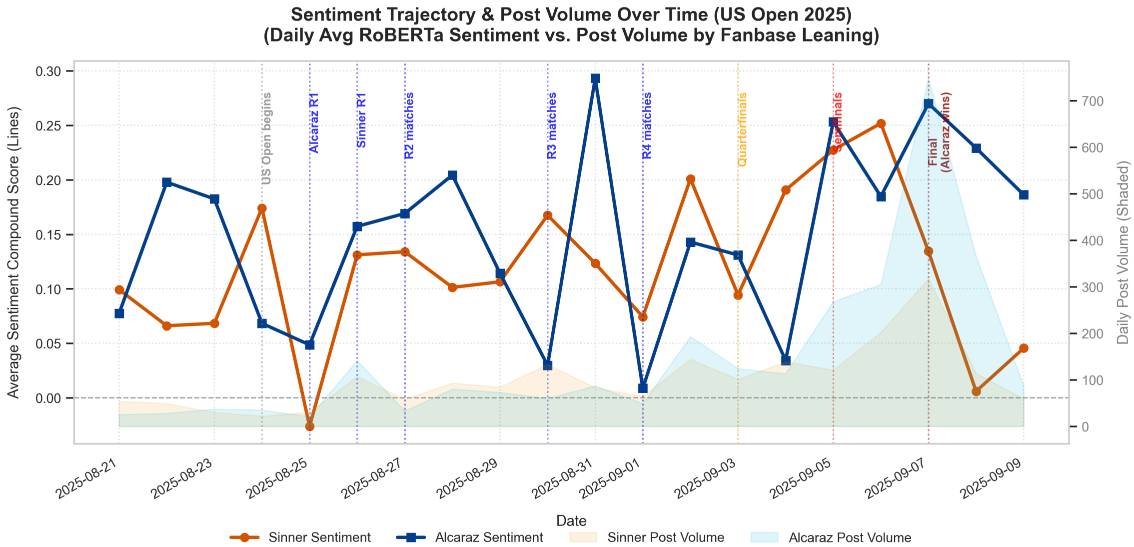


Figure 4: Daily average RoBERTa sentiment for Sinner vs. Alcaraz fans (lines) against daily post volume (shaded), annotated with US Open milestones. Sentiment tracks the tournament and diverges by fanbase around the final.

backends. Figure **??** compares dominant-emotion shares between the NRC lexicon and GoEmotions (BERT). Both agree that most posts are emotionally *neutral*, but the neural backend is far more conservative (neutral ≈ 0.84 vs. NRC ≈ 0.57). NRC surfaces more *anticipation*, *fear* and *trust*, whereas BERT distributes the remaining mass mainly across *trust*, *anticipation* and *joy*. Crucially, both backends agree that negative emotions (anger, disgust) are rare—consistent with the sentiment donut.

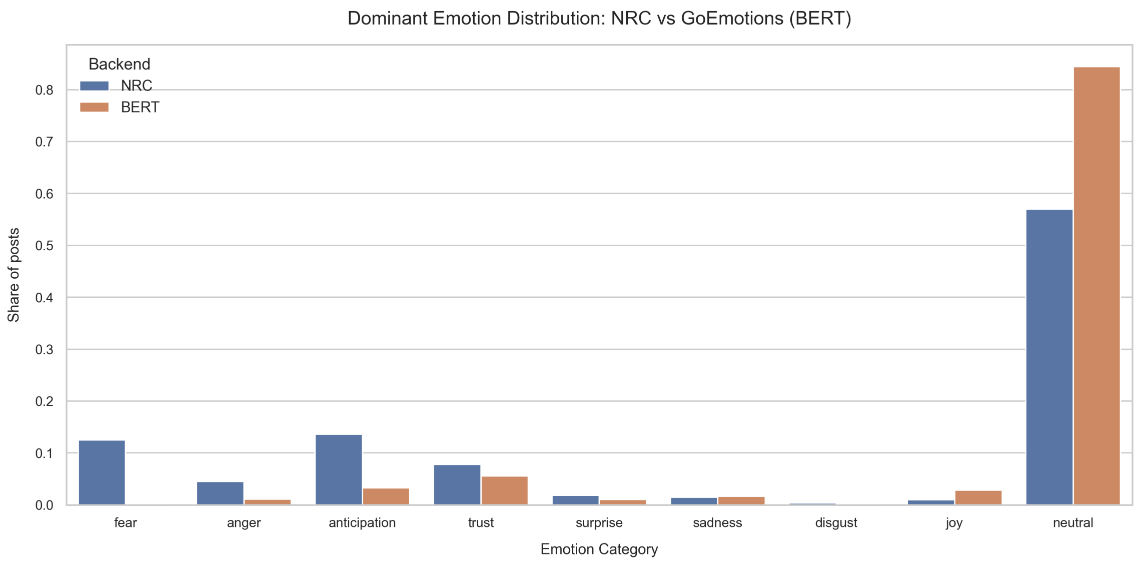
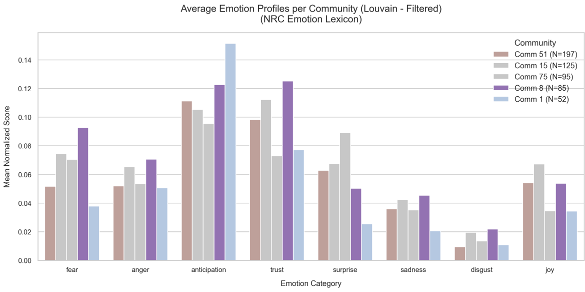
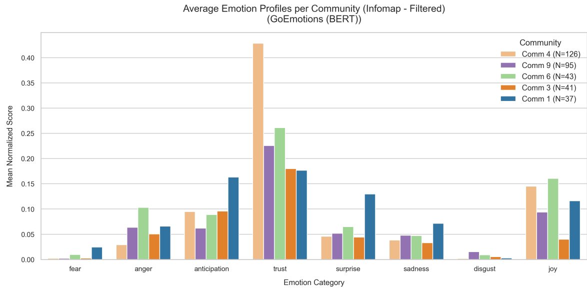


Figure 5: Dominant emotion distribution: NRC lexicon vs. GoEmotions (BERT). The neural backend labels more posts neutral; the lexicon assigns more anticipation/fear/trust.

Community emotion profiles. Per-community emotion profiles (Figure **??**) show that the largest Louvain communities share a positive affective signature—*anticipation* and *trust* dominate, with *joy* present and negative emotions subdued—but differ in magnitude, reflecting varying engagement intensity. Profiles are generated for both community algorithms (Louvain, Infomap) and both emotion backends (NRC, BERT).



(a) Louvain communities, NRC lexicon.



(b) Infomap communities, GoEmotions (BERT).

Figure 6: Average emotion profiles for the top communities. Anticipation and trust dominate across clusters; negative emotions stay low.

4.3 Stance & polarisation

The four-panel stance summary (Figure **??**) gives the headline picture. Of **2,296 analysed users**, most are **neutral (1,035)**, followed by **Alcaraz (835)** and **Sinner (426)**—a tilt towards Alcaraz consistent with his final victory. Net stance scores are tightly concentrated around zero (score SD $= 0.327$), and the polarisation diagnostics are mild: stance assortativity is slightly *negative* (-0.071) and the cross-stance edge ratio is 0.147. In other words, despite the high *structural* modularity of the graph, users are *not* strongly segregated by stance— fans of opposing players still interact (often via shared news hubs), so this is not a classic echo chamber.

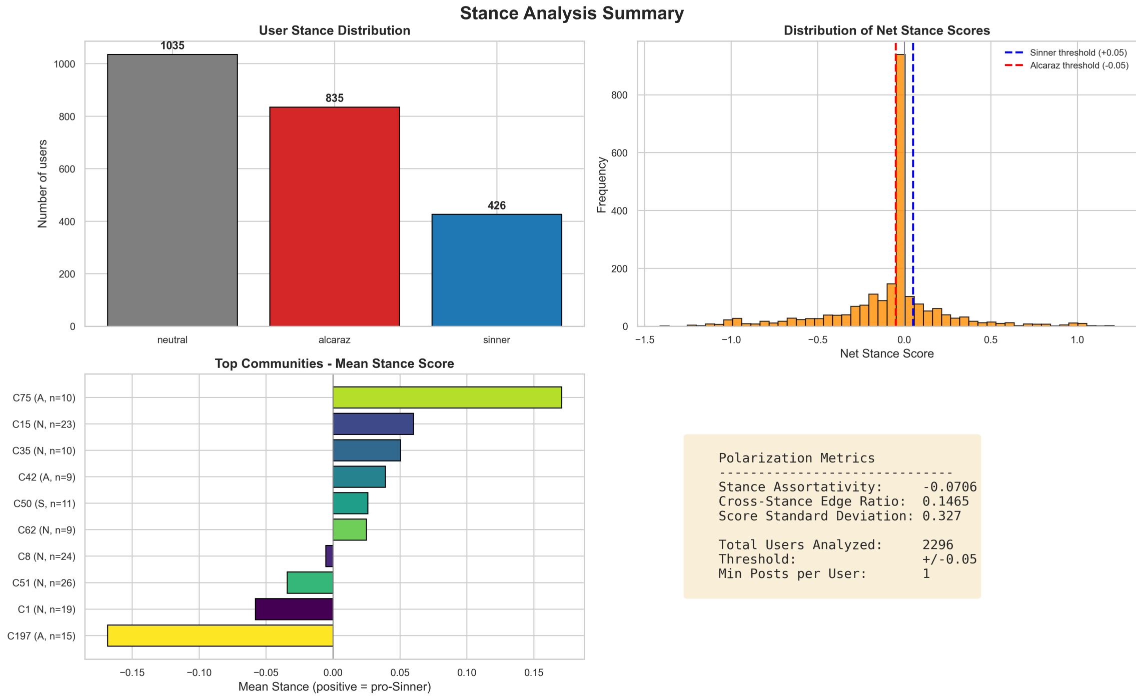


Figure 7: Stance analysis summary. Top-left: user stance counts (neutral 1035, Alcaraz 835, Sinner 426). Top-right: net-stance score histogram with ± 0.05 thresholds. Bottom-left: mean stance of top communities. Bottom-right: polarisation metrics (assortativity -0.071, cross-stance edge ratio 0.147, score SD 0.327).

Sentiment-aware vs. frequency baseline. Figure ?? contrasts a frequency-only baseline with the refined, sentiment-aware leaning. Folding RoBERTa sentiment and GoEmotions sentiment into the stance moves users between classes: the Sinner camp shrinks from **454 to 426**, Alcaraz edges up from **814 to 835**, and neutral grows from **1,028 to 1,035**. The reclassified users are typically those who *mention* a player frequently but *express* sentiment inconsistent with that mention pattern (e.g. criticising the player they name most)—exactly the cases a raw mention count mislabels. This demonstrates the value of fusing sentiment with frequency rather than counting mentions alone.

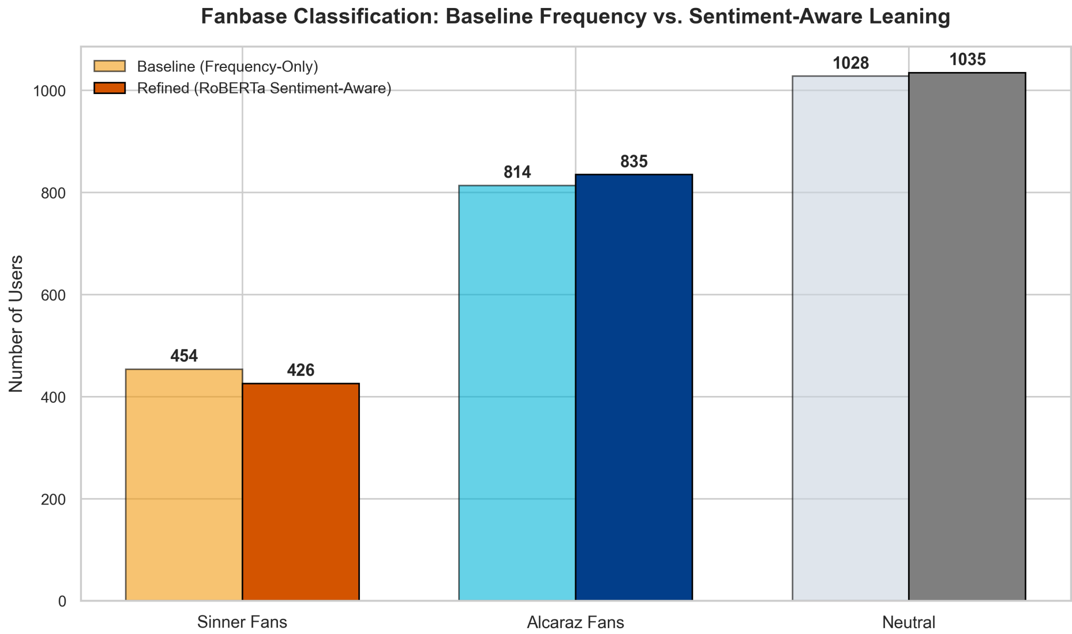


Figure 8: Fanbase classification: frequency-only baseline vs. sentiment-aware refinement. Sentiment reclassifies borderline users, shrinking the Sinner camp and growing the neutral pool.

Community vocabulary. A word cloud over the whole community (Figure ??) is dominated by the players' names and tournament/tennis vocabulary, confirming that the crawl captured on-topic discourse.

Figure 9: Word cloud of the community corpus—player names and US Open/tennis vocabulary dominate.

5 Discussion

5.1 Interpretation

The analysis paints a picture of a **vibrant, on-topic, but only mildly polarised** online community:

- **Structurally fragmented, not opinion-segregated.** The graph is extremely modular (Louvain $Q \approx 0.97$) yet stance assortativity is near zero. Clustering is driven by *who replies to whom* (frequently shared news/aggregator hubs) more than by partisan allegiance.
- **Positive, low-toxicity tone.** Most posts are neutral; among sentiment-bearing posts, positive outweighs negative $\approx$ 2.5:1, and negative emotions (anger, disgust) are rare under both emotion backends.
- **Sentiment tracks the tournament.** The time series reacts to rounds and diverges by fanbase around the final, illustrating how match outcomes drive emotional response.
- **Sentiment-aware stance matters.** Refining a frequency baseline with sentiment reclassifies a meaningful slice of users, correcting cases where mention frequency and expressed sentiment disagree.

5.2 Strengths

- **Reproducible DAG** with a frozen intermediate dataset, so any downstream stage can be re-run without recomputing the expensive NLP enrichment (seeds fixed at 42).
- **Two emotion backends** (lexicon vs. neural) provide robustness and expose how sensitive emotion attribution is to the scoring method.
- **Fine-grained stance** via sentence-level splitting of dual-mention posts.

5.3 Limitations & future work

- **Recall.** The Bluesky search API does not guarantee exhaustive retrieval; posts without the chosen keywords are missed.
- **Sarcasm & irony.** RoBERTa sentiment can misread sarcastic praise/criticism; sarcasm detection is a natural extension.
- **Weight calibration.** The stance weights (0.50 sentiment, 0.15 frequency, and 0.35 emotion) are heuristic; a data-driven fit against labelled stances would improve accuracy and optimize weight distribution.
- **Static communities.** Community detection is computed once over the whole window; a temporal (sliding-window) analysis could reveal how clusters form and dissolve around matches.

5.4 Conclusion

To give a clear answer to our initial question: the Sinner and Alcaraz fan communities on Bluesky manifest through a dual landscape characterized by structural fragmentation and emotional cohesion. Structurally, the interaction network is sparse (density ≈ 0.002) but highly modular ($Q \approx 0.97$), meaning fans cluster in separate discussion groups. However, this division is not driven by partisan echo chambers: stance assortativity is near zero (-0.071), and users primarily cluster around central, neutral sports news hubs which act as bridges between rival fan groups. Emotionally, the discourse is predominantly civil and positive, with neutral and encouraging posts vastly outnumbering hostile ones. This emotional response is highly event-driven, as codebase fanbase sentiment trajectories directly react to and diverge around key match outcomes like the tournament final. Lastly, our study demonstrates that a sentiment-aware stance classifier is methodologically essential to correctly identify user leanings compared to raw mention frequencies.

By integrating social-network analysis, NLP-based sentiment/emotion mining and sentiment-aware stance detection, this project characterises the Bluesky discourse around the Sinner–Alcaraz rivalry at the 2025 US Open. The pipeline is modular and reproducible, and readily adapts to other rivalries or platforms.